
EEE 6778. Applied Machine Learning II



Module 6. Edge AI

Definitions

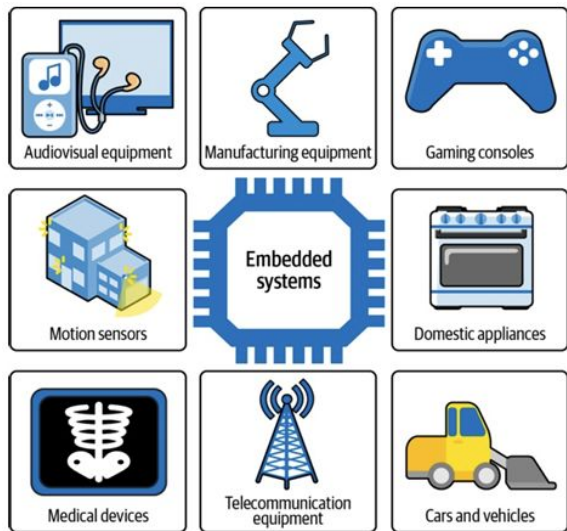
The Hardware of Edge AI

The Software for Edge AI



Definitions

What is Edge AI?



(Situnayake and Plunkett, 2023)



AI

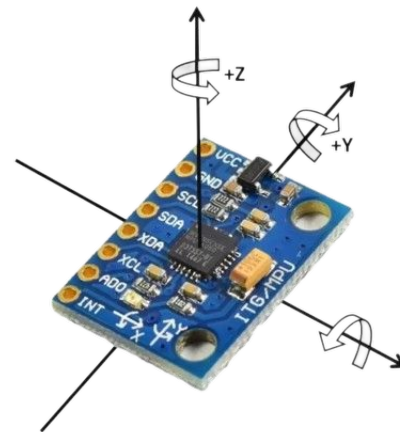
The combination between edge devices (embedded systems) and Artificial Intelligence

What is Edge AI?

IoT	AIoT	Edge AI – Edge Intelligence
The Internet of Things (IoT) is a network of physical objects, like appliances and devices, embedded with sensors and connected to the internet.	AIoT, or Artificial Intelligence of Things, merges Artificial Intelligence (AI) with the Internet of Things (IoT). It enables IoT devices to collect data and make smart decisions using AI algorithms	Edge AI, or Edge Artificial Intelligence, involves running AI algorithms and data processing on local devices, close to data sources rather than relying solely on distant cloud servers.

The Nature of Sensor Data

- Edge devices bridge the physical and digital world using sensors
- These devices generate large volumes of high-frequency data
- Data from sensors is high in volume but low in immediate informational value
- Traditionally, data is sent to central servers for processing →
Transmitting all sensor data is expensive and energy-intensive

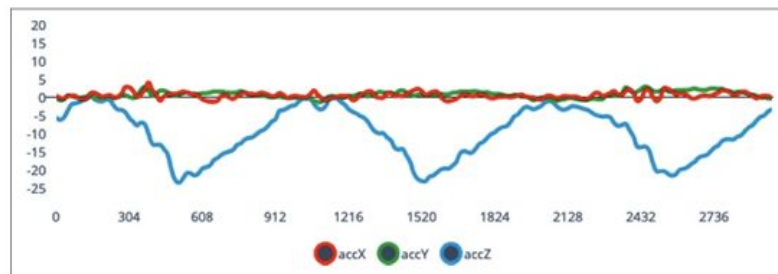


Example: Accelerometer generates many readings per second, but insights emerge only in aggregate

The Nature of Sensor Data

- Analyzing data by hand can be tricky, and it generally requires expert knowledge about the domain (such as human movements during sport).

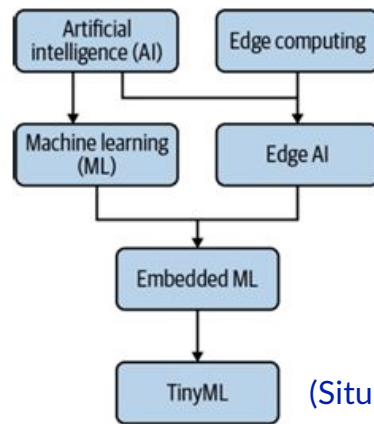
This is where Deep Learning becomes very handy!



(Situnayake and Plunkett, 2023)

- Edge AI processes data locally—on the device where it is generated
- Reduces dependency on connectivity and central servers
- Only important information needs to be transmitted
- Saves energy, reduces costs, and enables faster decisions

What are the differences between Embedded ML, TinyML, Edge AI, Edge Intelligence?



(Situnayake and Plunkett, 2023)

Why Do We Need Edge AI?

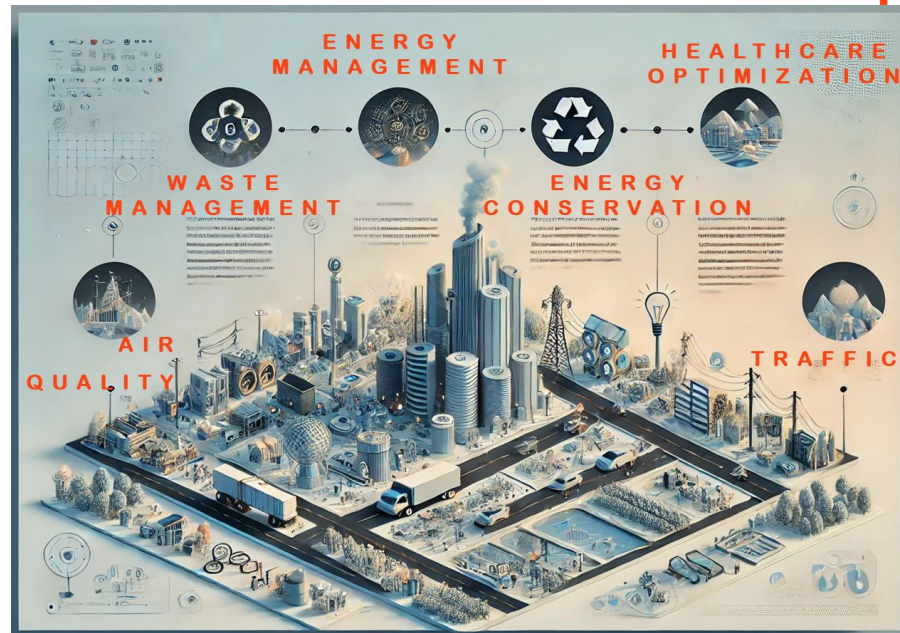
- Jeff Bier, founder of the Edge AI and Vision Alliance, created a tool for expressing the benefits of edge AI. It consists of five words:
 - **B**andwidth
 - **L**atency
 - **E**conomics
 - **R**eliability
 - **P**rivacy
- There doesn't have to be a strong argument for every word of the acronym: even meeting just one or two criteria, if compelling enough, can give merit to a use case.

Why Do We Need Edge AI?

- **Bandwidth:** Edge AI reduces the need to transmit high volumes of raw sensor data over limited or costly bandwidth connections.
- **Latency:** Edge AI eliminates round-trip communication delays, enabling real-time responsiveness
- **Economics:** Edge AI reduces the cost of connectivity and cloud infrastructure by minimizing data transmission
- **Reliability:** On-device AI increases system reliability by reducing dependence on network connections
- **Privacy:** Edge AI protects user privacy by keeping data processing local, minimizing the risk of exposing sensitive information through cloud storage or transmission.

Edge AI for Good

- The unique benefits of edge AI provide a new set of tools that can be applied to some of our world's biggest problems.
- Technologists in areas like conservation, healthcare, and education are already using edge AI to make a big impact



AI Image Generated
by DALL-E



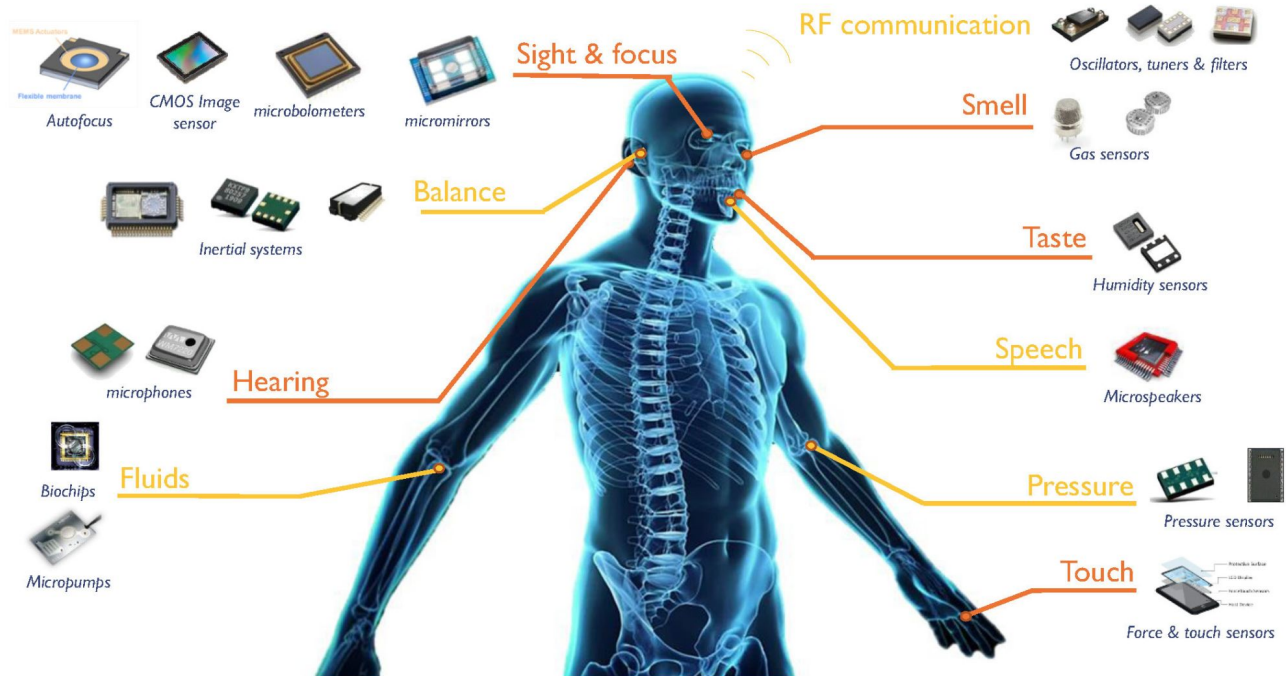
The Hardware of Edge AI

Sensors, Signals and Sources of Data

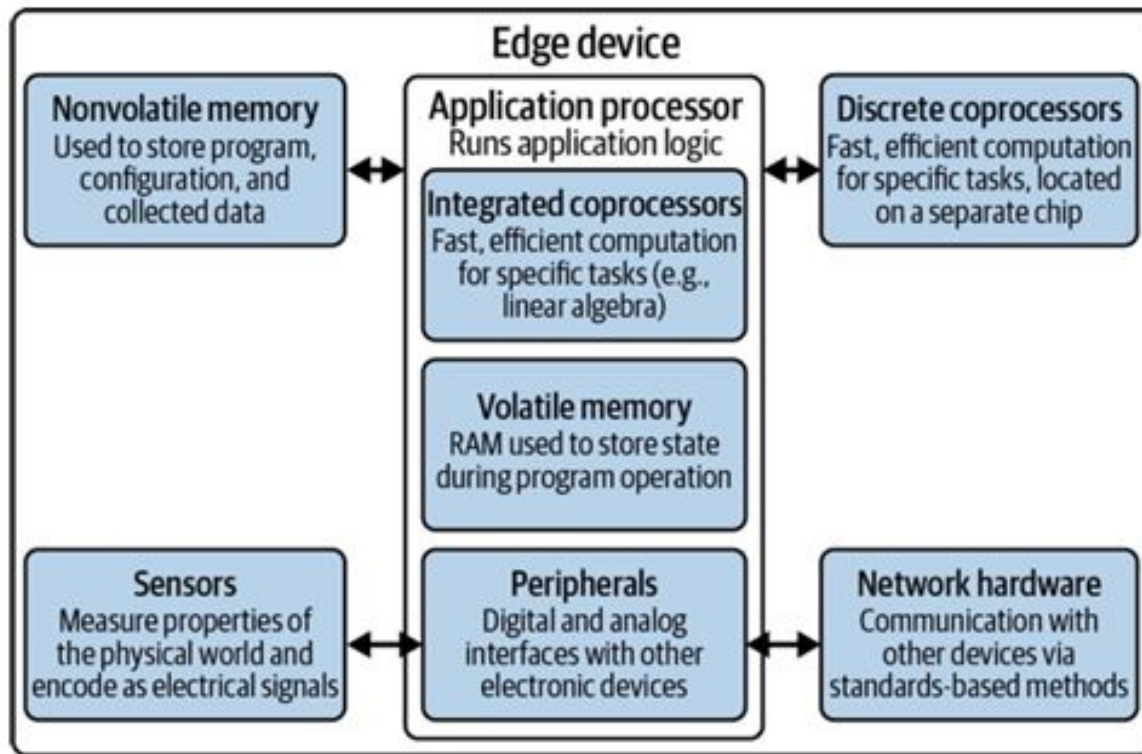
- Different sensors provide data in different formats
 - A **time series** may contain multiple values from the same physical sensor—e.g., a single sensor component may provide readings of both temperature and humidity.
 - A special case of time series data, **audio signals** represent the oscillation of sound waves as they travel through the air. Captured at very high frequency!
 - **Images data and video** represent the measurements taken by a sensor that captures an entire scene, as opposed to a single point. Some sensors, like **cameras**, use an array of tiny elements to capture data from the entire scene in one go. Other sensors, like **LIDAR** (Light Detection and Ranging), build up an image by mechanically sweeping a single sensor element across the scene over a period of time.

Sensors and Human Senses

(Source: Sensors for Wearable Electronics & Mobile Healthcare, Yole Développement, June 2015)



Processors for AI

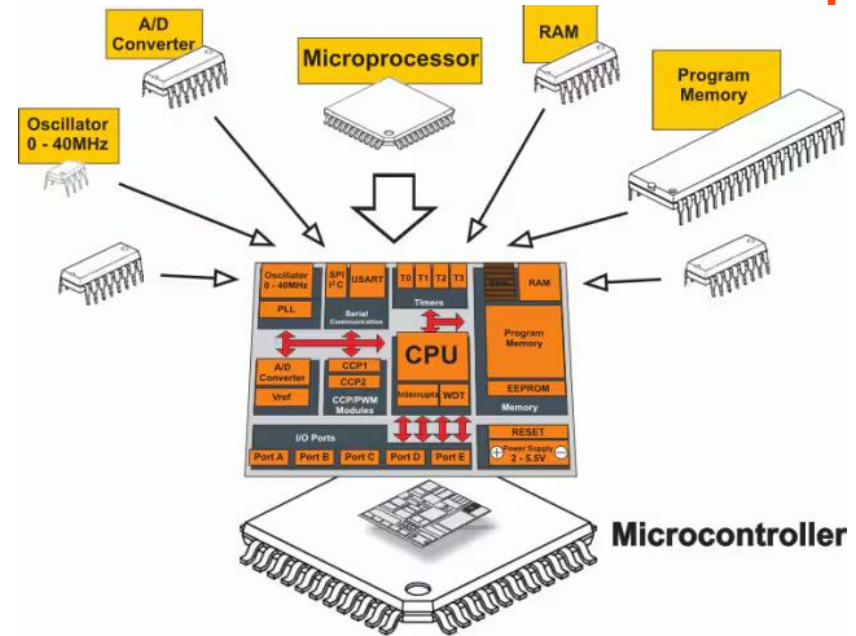


(Situnayake and Plunkett, 2023)

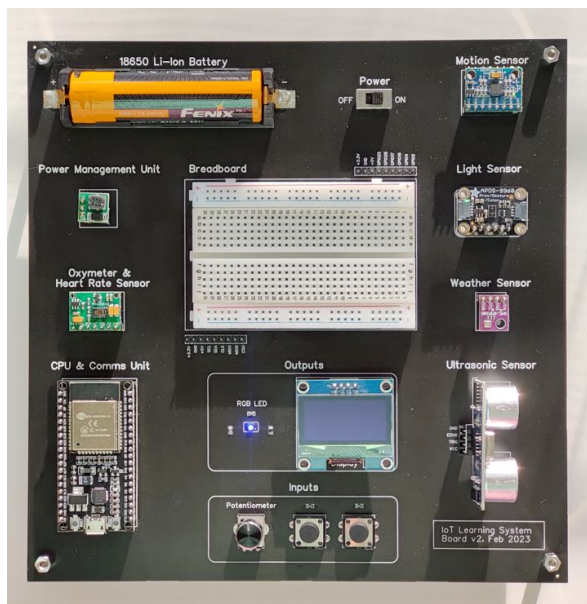
Architecture of an edge device

Microcontrollers (MCU)

- A microcontroller is a compact integrated circuit (IC) that contains a processor, memory, and input/output peripherals, designed to perform specific control tasks.
- It's crucial for Edge AI because it enables low-power, real-time processing directly on devices,



Development Boards and MCUs



- MCUs, require integration with power supplies, I/O interfaces, and sensors to operate as part of a complete edge AI device—typically achieved through custom PCBs in production systems.
- For early development, engineers use development boards that package MCUs or more powerful processors with all necessary peripherals, allowing rapid prototyping

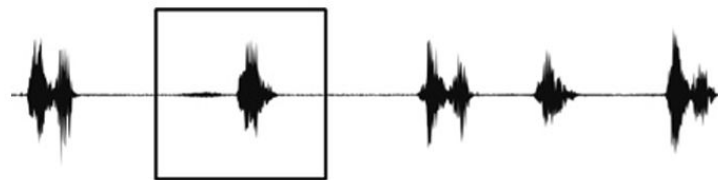
AHA Board – EdTech to teach Edge Intelligence and Applications



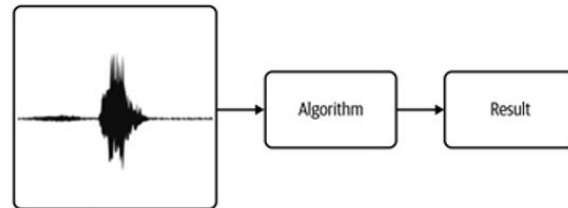
The Software for Edge AI

Feature Engineering

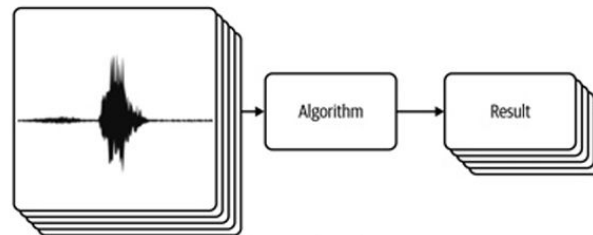
- The majority of sensors produce time series data.
- The goal of an edge AI application is to take these streams of time series data and make sense of them.
- The most common way to manage streams is to chop a time series into chunks, often called windows, then analyze the chunks one at a time



1. A window is taken from a stream of data.



2. The window is processed by an algorithm.

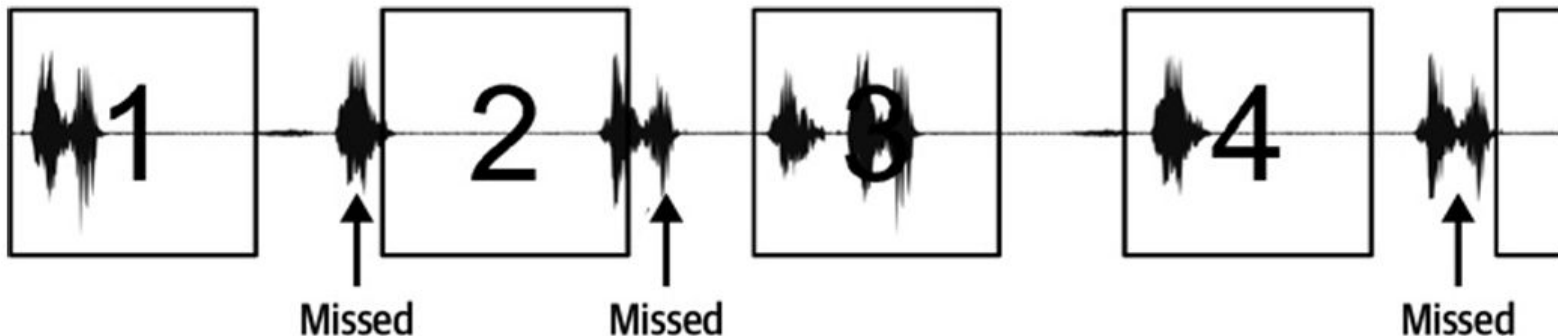


3. A series of windows produces a series of results.

(Situnayake and Plunkett, 2023)

The problem of latency

- If the frame rate is too low, some parts of the signal will not be processed;
 - if you are trying to detect short-lived events, this might mean that some events are missed → **The choice of the window is very important!**



(Situnayake and Plunkett, 2023)

Digital Signal Processing Algorithms

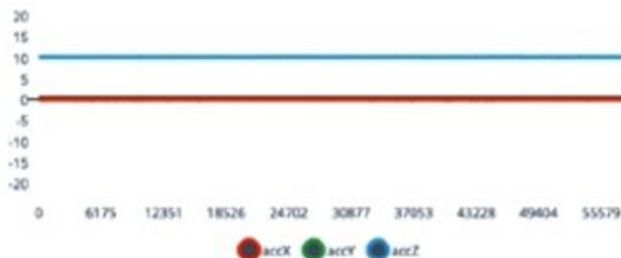
- There are hundreds of different signal processing algorithms that can help digest the signals produced by sensors. Examples:
 - Resampling (Upsampling or downsampling)
 - Filtering
 - Spectral Analysis
 - Image Feature Detection

Machine Learning, Deep Learning and Edge AI

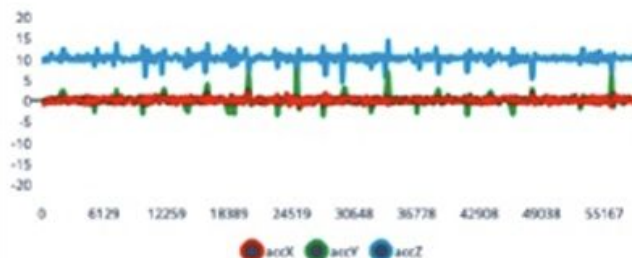
Possible labels: [idle, moving]



idle: 96%, moving: 4%



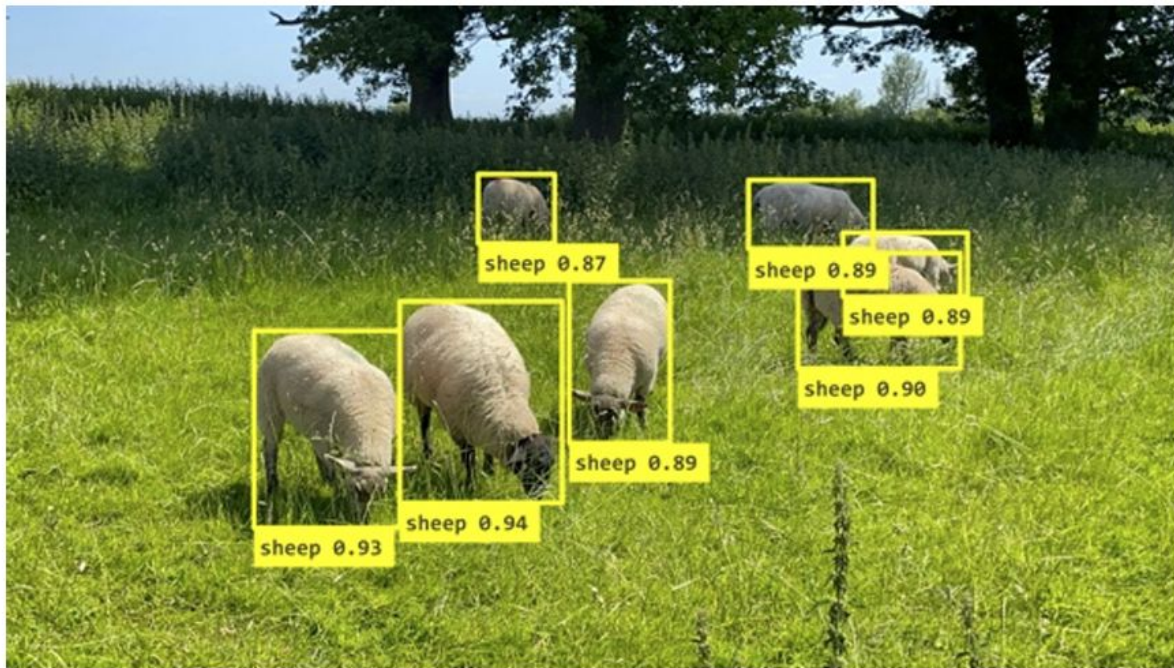
idle: 2%, moving: 98%



(Situnayake and Plunkett, 2023)

Classification

Machine Learning, Deep Learning and Edge AI



(Situnayake and Plunkett, 2023)

Regression

Machine Learning, Deep Learning and Edge AI

- Plus
 - Anomaly detection
 - Clustering
 - Dimensionality Reduction
 - Transformation
 - etc.

**When to use Classical
Machine Learning vs.
Deep Learning?**

Optimization for Edge Devices

- **Algorithm selection:** Choose simpler models or ones supported by hardware accelerators.
- **Feature reduction:** Minimize data and complexity before feeding into models.
- **Model compression and optimization:**
 - Quantization: Lowering numerical precision (e.g., 32-bit to 8-bit) to reduce size and increase speed.
 - Pruning: Remove unimportant weights to make sparse models.
 - Knowledge Distillation: Train a small model to mimic a large one.